

```
4  #include <string>
5
6  using namespace std;
7
8  class BankAccount {
9  private:
10     double balance;
11
12
13     void saveTransaction(const string& type, const string& item, double amount) {
14         ofstream file("transactions.txt", ios::app);
15         if (!file) {
16             cerr << "[ERROR] Could not open transactions.txt\n";
17             return;
18         }
19         file << fixed << setprecision(2);
20         file << type;
21         if (!item.empty()) file << " - " << item;
22         file << ": $" << amount << " | Balance: $" << balance << '\n';
23         file.close();
24     }
25
26 public:
27
28     BankAccount() : balance(0.0) {}
29     explicit BankAccount(double starting) : balance(starting) {
30
31         saveTransaction("Start", "", balance);
32     }
33
34
35     void deposit(double amount) {
36         if (amount <= 0) {
37             cout << "Deposit must be positive.\n";
38             return;
39         }
40         balance += amount;
41         cout << "Deposited $" << fixed << setprecision(2) << amount
42             << ". New balance: $" << balance << '\n';
43         saveTransaction("Deposit", "", amount);
44     }
45
46
47     void makePurchase(const string& item, double cost) {
48         if (cost <= 0) {
49             cout << "Purchase amount must be positive.\n";
50             return;
51         }
52         if (cost > balance) {
53             cout << "Insufficient funds for \"" << item << "\" (cost $"
54                 << fixed << setprecision(2) << cost << ").\n";
55             return;
56         }
57         balance -= cost;
58         cout << "Purchased \"" << item << "\" for $" << fixed << setprecision(2)
59             << cost << ". New balance: $" << balance << '\n';
60         saveTransaction("Purchase", item, cost);
61     }
62
63
64     void displayBalance() const {
65         cout << "Current Balance: $" << fixed << setprecision(2) << balance << '\n';
66     }
67 };
68
69 int main() {
70     ofstream reset("transactions.txt");
71     reset << "=== Transaction Log ===\n";
72     reset.close();
73
74     BankAccount myAccount;
75     myAccount.deposit(100.00);
76     myAccount.makePurchase("Coffee", 4.50);
77     myAccount.makePurchase("Book", 25.00);
78     myAccount.displayBalance();
79
80     myAccount.makePurchase("Laptop", 2000.00);
81
82     return 0;
83 }
84
```


→ **CPP Class** ./midterm

Deposited \$100.00. New balance: \$100.00

Purchased "Coffee" for \$4.50. New balance: \$95.50

Purchased "Book" for \$25.00. New balance: \$70.50

Current Balance: \$70.50

Insufficient funds for "Laptop" (cost \$2000.00).

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